

Sales trend with seasonal fluctuations.

Solution to Practice Problem

Construct X and β so that the k th row of $X\beta$ is the predicted y -value that corresponds to the data point (x_k, y_k) , namely

$$\beta_0 + \beta_1 x_k + \beta_2 \sin(2\pi x_k/12)$$

It should be clear that

$$X = \begin{bmatrix} 1 & x_1 & \sin(2\pi x_1/12) \\ \vdots & \vdots & \vdots \\ 1 & x_n & \sin(2\pi x_n/12) \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

6.7 Inner Product Spaces

Notions of length, distance, and orthogonality are often important in applications involving a vector space. For $\mathbb{R}^n$, these concepts were based on the properties of the inner product listed in Theorem 1 of Section 6.1. For other spaces, we need analogues of the inner product with the same properties. The conclusions of Theorem 1 now become *axioms* in the following definition.

DEFINITION

An **inner product** on a vector space V is a function that, to each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V , associates a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ and satisfies the following axioms, for all $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ in V and all scalars c :

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
3. $\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$
4. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$

A vector space with an inner product is called an **inner product space**.

The vector space $\mathbb{R}^n$ with the standard inner product is an inner product space, and nearly everything discussed in this chapter for $\mathbb{R}^n$ carries over to inner product spaces. The examples in this section and the next lay the foundation for a variety of applications treated in courses in engineering, physics, mathematics, and statistics.

EXAMPLE 1 Fix any two positive numbers—say, 4 and 5—and for vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ in $\mathbb{R}^2$, set

$$\langle \mathbf{u}, \mathbf{v} \rangle = 4u_1v_1 + 5u_2v_2 \quad (1)$$

Show that equation (1) defines an inner product.

SOLUTION Certainly Axiom 1 is satisfied, because $\langle \mathbf{u}, \mathbf{v} \rangle = 4u_1v_1 + 5u_2v_2 = 4v_1u_1 + 5v_2u_2 = \langle \mathbf{v}, \mathbf{u} \rangle$. If $\mathbf{w} = (w_1, w_2)$, then

$$\begin{aligned} \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle &= 4(u_1 + v_1)w_1 + 5(u_2 + v_2)w_2 \\ &= 4u_1w_1 + 5u_2w_2 + 4v_1w_1 + 5v_2w_2 \\ &= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle \end{aligned}$$

This verifies Axiom 2. For Axiom 3, compute

$$\langle c\mathbf{u}, \mathbf{v} \rangle = 4(cu_1)v_1 + 5(cu_2)v_2 = c(4u_1v_1 + 5u_2v_2) = c\langle \mathbf{u}, \mathbf{v} \rangle$$

For Axiom 4, note that $\langle \mathbf{u}, \mathbf{u} \rangle = 4u_1^2 + 5u_2^2 \geq 0$, and $4u_1^2 + 5u_2^2 = 0$ only if $u_1 = u_2 = 0$, that is, if $\mathbf{u} = \mathbf{0}$. Also, $\langle \mathbf{0}, \mathbf{0} \rangle = 0$. So (1) defines an inner product on $\mathbb{R}^2$. ■

Inner products similar to (1) can be defined on $\mathbb{R}^n$. They arise naturally in connection with “weighted least-squares” problems, in which weights are assigned to the various entries in the sum for the inner product in such a way that more importance is given to the more reliable measurements.

From now on, when an inner product space involves polynomials or other functions, we will write the functions in the familiar way, rather than use the boldface type for vectors. Nevertheless, it is important to remember that each function *is* a vector when it is treated as an element of a vector space.

EXAMPLE 2 Let $t_0, \dots, t_n$ be distinct real numbers. For p and q in $\mathbb{P}_n$, define

$$\langle p, q \rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + \cdots + p(t_n)q(t_n) \quad (2)$$

Inner product Axioms 1–3 are readily checked. For Axiom 4, note that

$$\langle p, p \rangle = [p(t_0)]^2 + [p(t_1)]^2 + \cdots + [p(t_n)]^2 \geq 0$$

Also, $\langle \mathbf{0}, \mathbf{0} \rangle = 0$. (The boldface zero here denotes the zero polynomial, the zero vector in $\mathbb{P}_n$.) If $\langle p, p \rangle = 0$, then p must vanish at $n + 1$ points: $t_0, \dots, t_n$. This is possible only if p is the zero polynomial, because the degree of p is less than $n + 1$. Thus (2) defines an inner product on $\mathbb{P}_n$. ■

EXAMPLE 3 Let V be $\mathbb{P}_2$, with the inner product from Example 2, where $t_0 = 0$, $t_1 = \frac{1}{2}$, and $t_2 = 1$. Let $p(t) = 12t^2$ and $q(t) = 2t - 1$. Compute $\langle p, q \rangle$ and $\langle q, q \rangle$.

SOLUTION

$$\begin{aligned} \langle p, q \rangle &= p(0)q(0) + p\left(\frac{1}{2}\right)q\left(\frac{1}{2}\right) + p(1)q(1) \\ &= (0)(-1) + (3)(0) + (12)(1) = 12 \\ \langle q, q \rangle &= [q(0)]^2 + \left[q\left(\frac{1}{2}\right)\right]^2 + [q(1)]^2 \\ &= (-1)^2 + (0)^2 + (1)^2 = 2 \end{aligned} \quad \blacksquare$$

Lengths, Distances, and Orthogonality

Let V be an inner product space, with the inner product denoted by $\langle \mathbf{u}, \mathbf{v} \rangle$. Just as in $\mathbb{R}^n$, we define the **length**, or **norm**, of a vector $\mathbf{v}$ to be the scalar

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$$

Equivalently, $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle$. (This definition makes sense because $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, but the definition *does not* say that $\langle \mathbf{v}, \mathbf{v} \rangle$ is a “sum of squares,” because $\mathbf{v}$ need not be an element of $\mathbb{R}^n$.)

A **unit vector** is one whose length is 1. The **distance between $\mathbf{u}$ and $\mathbf{v}$** is $\|\mathbf{u} - \mathbf{v}\|$. Vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

EXAMPLE 4 Let $\mathbb{P}_2$ have the inner product (2) of Example 3. Compute the lengths of the vectors $p(t) = 12t^2$ and $q(t) = 2t - 1$.

SOLUTION

$$\begin{aligned}\|p\|^2 &= \langle p, p \rangle = [p(0)]^2 + \left[p\left(\frac{1}{2}\right)\right]^2 + [p(1)]^2 \\ &= 0 + [3]^2 + [12]^2 = 153 \\ \|p\| &= \sqrt{153}\end{aligned}$$

From Example 3, $\langle q, q \rangle = 2$. Hence $\|q\| = \sqrt{2}$. ■

The Gram–Schmidt Process

The existence of orthogonal bases for finite-dimensional subspaces of an inner product space can be established by the Gram–Schmidt process, just as in $\mathbb{R}^n$. Certain orthogonal bases that arise frequently in applications can be constructed by this process.

The orthogonal projection of a vector onto a subspace W with an orthogonal basis can be constructed as usual. The projection does not depend on the choice of orthogonal basis, and it has the properties described in the Orthogonal Decomposition Theorem and the Best Approximation Theorem.

EXAMPLE 5 Let V be $\mathbb{P}_4$ with the inner product in Example 2, involving evaluation of polynomials at $-2, -1, 0, 1$, and 2 , and view $\mathbb{P}_2$ as a subspace of V . Produce an orthogonal basis for $\mathbb{P}_2$ by applying the Gram–Schmidt process to the polynomials $1, t$, and t^2 .

SOLUTION The inner product depends only on the values of a polynomial at $-2, \dots, 2$, so we list the values of each polynomial as a vector in $\mathbb{R}^5$, underneath the name of the polynomial:¹

$$\begin{array}{lcl} \text{Polynomial:} & 1 & t & t^2 \\ \text{Vector of values:} & \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, & \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, & \begin{bmatrix} 4 \\ 1 \\ 0 \\ 1 \\ 4 \end{bmatrix} \end{array}$$

The inner product of two polynomials in V equals the (standard) inner product of their corresponding vectors in $\mathbb{R}^5$. Observe that t is orthogonal to the constant function 1 . So take $p_0(t) = 1$ and $p_1(t) = t$. For p_2 , use the vectors in $\mathbb{R}^5$ to compute the projection of t^2 onto $\text{Span}\{p_0, p_1\}$:

$$\begin{aligned}\langle t^2, p_0 \rangle &= \langle t^2, 1 \rangle = 4 + 1 + 0 + 1 + 4 = 10 \\ \langle p_0, p_0 \rangle &= 5 \\ \langle t^2, p_1 \rangle &= \langle t^2, t \rangle = -8 + (-1) + 0 + 1 + 8 = 0\end{aligned}$$

The orthogonal projection of t^2 onto $\text{Span}\{1, t\}$ is $\frac{10}{5}p_0 + 0p_1$. Thus

$$p_2(t) = t^2 - 2p_0(t) = t^2 - 2$$

¹ Each polynomial in $\mathbb{P}_4$ is uniquely determined by its value at the five numbers $-2, \dots, 2$. In fact, the correspondence between p and its vector of values is an isomorphism, that is, a one-to-one mapping onto $\mathbb{R}^5$ that preserves linear combinations.

An orthogonal basis for the subspace $\mathbb{P}_2$ of V is

$$\begin{array}{l} \text{Polynomial} \\ \text{Vector of values} \end{array} \quad \begin{array}{c} p_0 \\ p_1 \\ p_2 \end{array} \quad \begin{array}{c} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \\ \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \\ \begin{bmatrix} 2 \\ -1 \\ -2 \\ -1 \\ 2 \end{bmatrix} \end{array} \quad (3)$$

■

Best Approximation in Inner Product Spaces

A common problem in applied mathematics involves a vector space V whose elements are functions. The problem is to approximate a function f in V by a function g from a specified subspace W of V . The “closeness” of the approximation of f depends on the way $\|f - g\|$ is defined. We will consider only the case in which the distance between f and g is determined by an inner product. In this case, the *best approximation to f by functions in W* is the orthogonal projection of f onto the subspace W .

EXAMPLE 6 Let V be $\mathbb{P}_4$ with the inner product in Example 5, and let p_0 , p_1 , and p_2 be the orthogonal basis found in Example 5 for the subspace $\mathbb{P}_2$. Find the best approximation to $p(t) = 5 - \frac{1}{2}t^4$ by polynomials in $\mathbb{P}_2$.

SOLUTION The values of p_0 , p_1 , and p_2 at the numbers $-2, -1, 0, 1$, and 2 are listed in $\mathbb{R}^5$ vectors in (3) above. The corresponding values for p are $-3, 9/2, 5, 9/2$, and -3 . Compute

$$\begin{aligned} \langle p, p_0 \rangle &= 8, & \langle p, p_1 \rangle &= 0, & \langle p, p_2 \rangle &= -31 \\ \langle p_0, p_0 \rangle &= 5, & \langle p_2, p_2 \rangle &= 14 \end{aligned}$$

Then the best approximation in V to p by polynomials in $\mathbb{P}_2$ is

$$\begin{aligned} \hat{p} &= \text{proj}_{\mathbb{P}_2} p = \frac{\langle p, p_0 \rangle}{\langle p_0, p_0 \rangle} p_0 + \frac{\langle p, p_1 \rangle}{\langle p_1, p_1 \rangle} p_1 + \frac{\langle p, p_2 \rangle}{\langle p_2, p_2 \rangle} p_2 \\ &= \frac{8}{5} p_0 + \frac{-31}{14} p_2 = \frac{8}{5} - \frac{31}{14}(t^2 - 2). \end{aligned}$$

This polynomial is the closest to p of all polynomials in $\mathbb{P}_2$, when the distance between polynomials is measured only at $-2, -1, 0, 1$, and 2 . See Figure 1. ■

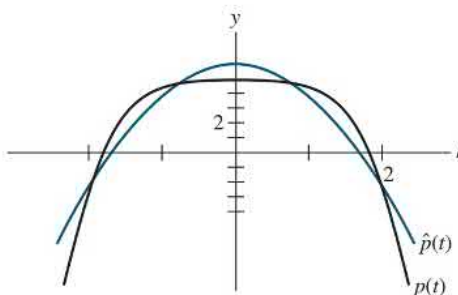


FIGURE 1

The polynomials p_0 , p_1 , and p_2 in Examples 5 and 6 belong to a class of polynomials that are referred to in statistics as *orthogonal polynomials*.² The orthogonality refers to the type of inner product described in Example 2.

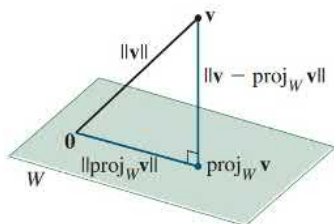


FIGURE 2
The hypotenuse is the longest side.

Two Inequalities

Given a vector $\mathbf{v}$ in an inner product space V and given a finite-dimensional subspace W , we may apply the Pythagorean Theorem to the orthogonal decomposition of $\mathbf{v}$ with respect to W and obtain

$$\|\mathbf{v}\|^2 = \|\text{proj}_W \mathbf{v}\|^2 + \|\mathbf{v} - \text{proj}_W \mathbf{v}\|^2$$

See Figure 2. In particular, this shows that the norm of the projection of $\mathbf{v}$ onto W does not exceed the norm of $\mathbf{v}$ itself. This simple observation leads to the following important inequality.

THEOREM 16

The Cauchy–Schwarz Inequality

For all $\mathbf{u}, \mathbf{v}$ in V ,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\| \quad (4)$$

PROOF If $\mathbf{u} = \mathbf{0}$, then both sides of (4) are zero, and hence the inequality is true in this case. (See Practice Problem 1.) If $\mathbf{u} \neq \mathbf{0}$, let W be the subspace spanned by $\mathbf{u}$. Recall that $\|c\mathbf{u}\| = |c| \|\mathbf{u}\|$ for any scalar c . Thus

$$\|\text{proj}_W \mathbf{v}\| = \left\| \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u} \right\| = \frac{|\langle \mathbf{v}, \mathbf{u} \rangle|}{|\langle \mathbf{u}, \mathbf{u} \rangle|} \|\mathbf{u}\| = \frac{|\langle \mathbf{v}, \mathbf{u} \rangle|}{\|\mathbf{u}\|^2} \|\mathbf{u}\| = \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|}{\|\mathbf{u}\|}$$

Since $\|\text{proj}_W \mathbf{v}\| \leq \|\mathbf{v}\|$, we have $\frac{|\langle \mathbf{u}, \mathbf{v} \rangle|}{\|\mathbf{u}\|} \leq \|\mathbf{v}\|$, which gives (4). ■

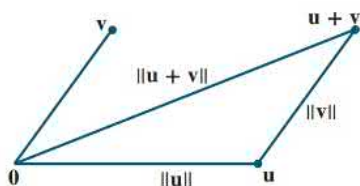


FIGURE 3
The lengths of the sides of a triangle.

The Cauchy–Schwarz inequality is useful in many branches of mathematics. A few simple applications are presented in the exercises. Our main need for this inequality here is to prove another fundamental inequality involving norms of vectors. See Figure 3.

THEOREM 17

The Triangle Inequality

For all $\mathbf{u}, \mathbf{v}$ in V ,

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

PROOF

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle + 2\langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &\leq \|\mathbf{u}\|^2 + 2|\langle \mathbf{u}, \mathbf{v} \rangle| + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| + \|\mathbf{v}\|^2 && \text{Cauchy–Schwarz} \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2 \end{aligned}$$

The triangle inequality follows immediately by taking square roots of both sides. ■

² See *Statistics and Experimental Design in Engineering and the Physical Sciences*, 2nd ed., by Norman L. Johnson and Fred C. Leone (New York: John Wiley & Sons, 1977). Tables there list “Orthogonal Polynomials,” which are simply the values of the polynomial at numbers such as $-2, -1, 0, 1$, and 2 .

An Inner Product for $C[a, b]$ (Calculus required)

Probably the most widely used inner product space for applications is the vector space $C[a, b]$ of all continuous functions on an interval $a \leq t \leq b$, with an inner product that we will describe.

We begin by considering a polynomial p and any integer n larger than or equal to the degree of p . Then p is in $\mathbb{P}_n$, and we may compute a “length” for p using the inner product of Example 2 involving evaluation at $n + 1$ points in $[a, b]$. However, this length of p captures the behavior at only those $n + 1$ points. Since p is in $\mathbb{P}_n$ for all large n , we could use a much larger n , with many more points for the “evaluation” inner product. See Figure 4.

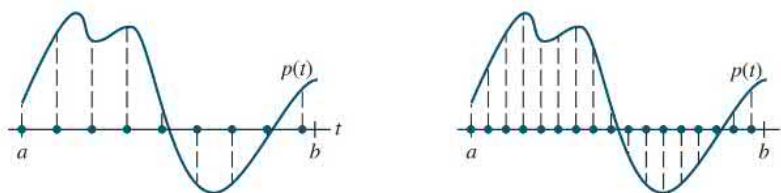
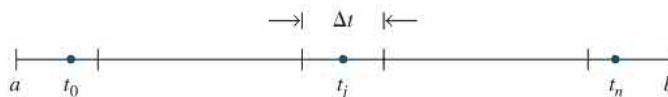


FIGURE 4 Using different numbers of evaluation points in $[a, b]$ to compute $\|p\|^2$.

Let us partition $[a, b]$ into $n + 1$ subintervals of length $\Delta t = (b - a)/(n + 1)$, and let $t_0, \dots, t_n$ be arbitrary points in these subintervals.



If n is large, the inner product on $\mathbb{P}_n$ determined by $t_0, \dots, t_n$ will tend to give a large value to $\langle p, p \rangle$, so we scale it down and divide by $n + 1$. Observe that $1/(n + 1) = \Delta t/(b - a)$, and define

$$\langle p, q \rangle = \frac{1}{n + 1} \sum_{j=0}^n p(t_j)q(t_j) = \frac{1}{b - a} \left[\sum_{j=0}^n p(t_j)q(t_j)\Delta t \right]$$

Now, let n increase without bound. Since polynomials p and q are continuous functions, the expression in brackets is a Riemann sum that approaches a definite integral, and we are led to consider the *average value* of $p(t)q(t)$ on the interval $[a, b]$:

$$\frac{1}{b - a} \int_a^b p(t)q(t) dt$$

This quantity is defined for polynomials of any degree (in fact, for all continuous functions), and it has all the properties of an inner product, as the next example shows. The scale factor $1/(b - a)$ is inessential and is often omitted for simplicity.

EXAMPLE 7 For f, g in $C[a, b]$, set

$$\langle f, g \rangle = \int_a^b f(t)g(t) dt \quad (5)$$

Show that (5) defines an inner product on $C[a, b]$.

SOLUTION Inner product Axioms 1–3 follow from elementary properties of definite integrals. For Axiom 4, observe that

$$\langle f, f \rangle = \int_a^b [f(t)]^2 dt \geq 0$$

The function $[f(t)]^2$ is continuous and nonnegative on $[a, b]$. If the definite integral of $[f(t)]^2$ is zero, then $[f(t)]^2$ must be identically zero on $[a, b]$, by a theorem in advanced calculus, in which case f is the zero function. Thus $\langle f, f \rangle = 0$ implies that f is the zero function on $[a, b]$. So (5) defines an inner product on $C[a, b]$. ■

EXAMPLE 8 Let V be the space $C[0, 1]$ with the inner product of Example 7, and let W be the subspace spanned by the polynomials $p_1(t) = 1$, $p_2(t) = 2t - 1$, and $p_3(t) = 12t^2$. Use the Gram–Schmidt process to find an orthogonal basis for W .

SOLUTION Let $q_1 = p_1$, and compute

$$\langle p_2, q_1 \rangle = \int_0^1 (2t - 1)(1) dt = (t^2 - t) \Big|_0^1 = 0$$

So p_2 is already orthogonal to q_1 , and we can take $q_2 = p_2$. For the projection of p_3 onto $W_2 = \text{Span}\{q_1, q_2\}$, compute

$$\begin{aligned} \langle p_3, q_1 \rangle &= \int_0^1 12t^2 \cdot 1 dt = 4t^3 \Big|_0^1 = 4 \\ \langle q_1, q_1 \rangle &= \int_0^1 1 \cdot 1 dt = t \Big|_0^1 = 1 \\ \langle p_3, q_2 \rangle &= \int_0^1 12t^2(2t - 1) dt = \int_0^1 (24t^3 - 12t^2) dt = 2 \\ \langle q_2, q_2 \rangle &= \int_0^1 (2t - 1)^2 dt = \frac{1}{6}(2t - 1)^3 \Big|_0^1 = \frac{1}{3} \end{aligned}$$

Then

$$\text{proj}_{W_2} p_3 = \frac{\langle p_3, q_1 \rangle}{\langle q_1, q_1 \rangle} q_1 + \frac{\langle p_3, q_2 \rangle}{\langle q_2, q_2 \rangle} q_2 = \frac{4}{1} q_1 + \frac{2}{1/3} q_2 = 4q_1 + 6q_2$$

and

$$q_3 = p_3 - \text{proj}_{W_2} p_3 = p_3 - 4q_1 - 6q_2$$

As a function, $q_3(t) = 12t^2 - 4 - 6(2t - 1) = 12t^2 - 12t + 2$. The orthogonal basis for the subspace W is $\{q_1, q_2, q_3\}$. ■

Practice Problems

Use the inner product axioms to verify the following statements.

1. $\langle \mathbf{v}, \mathbf{0} \rangle = \langle \mathbf{0}, \mathbf{v} \rangle = 0$.
2. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$.

6.7 Exercises

- Let $\mathbb{R}^2$ have the inner product of Example 1, and let $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (5, -1)$.
 - Find $\|\mathbf{x}\|$, $\|\mathbf{y}\|$, and $|\langle \mathbf{x}, \mathbf{y} \rangle|^2$.
 - Describe all vectors (z_1, z_2) that are orthogonal to $\mathbf{y}$.
- Let $\mathbb{R}^2$ have the inner product of Example 1. Show that the Cauchy–Schwarz inequality holds for $\mathbf{x} = (3, -2)$ and $\mathbf{y} = (-2, 1)$. [Suggestion: Study $|\langle \mathbf{x}, \mathbf{y} \rangle|^2$.]

Exercises 3–8 refer to $\mathbb{P}_2$ with the inner product given by evaluation at $-1, 0$, and 1 . (See Example 2.)

- Compute $\langle p, q \rangle$, where $p(t) = 4 + t$, $q(t) = 5 - 4t^2$.
- Compute $\langle p, q \rangle$, where $p(t) = 3t - t^2$, $q(t) = 3 + 2t^2$.
- Compute $\|p\|$ and $\|q\|$, for p and q in Exercise 3.
- Compute $\|p\|$ and $\|q\|$, for p and q in Exercise 4.
- Compute the orthogonal projection of q onto the subspace spanned by p , for p and q in Exercise 3.
- Compute the orthogonal projection of q onto the subspace spanned by p , for p and q in Exercise 4.
- Let $\mathbb{P}_3$ have the inner product given by evaluation at $-3, -1, 1$, and 3 . Let $p_0(t) = 1$, $p_1(t) = t$, and $p_2(t) = t^2$.
 - Compute the orthogonal projection of p_2 onto the subspace spanned by p_0 and p_1 .
 - Find a polynomial q that is orthogonal to p_0 and p_1 , such that $\{p_0, p_1, q\}$ is an orthogonal basis for $\text{Span}\{p_0, p_1, p_2\}$. Scale the polynomial q so that its vector of values at $(-3, -1, 1, 3)$ is $(1, -1, -1, 1)$.
- Let $\mathbb{P}_3$ have the inner product as in Exercise 9, with p_0, p_1 , and q the polynomials described there. Find the best approximation to $p(t) = t^3$ by polynomials in $\text{Span}\{p_0, p_1, q\}$.
- Let p_0, p_1 , and p_2 be the orthogonal polynomials described in Example 5, where the inner product on $\mathbb{P}_4$ is given by evaluation at $-2, -1, 0, 1$, and 2 . Find the orthogonal projection of t^3 onto $\text{Span}\{p_0, p_1, p_2\}$.
- Find a polynomial p_3 such that $\{p_0, p_1, p_2, p_3\}$ (see Exercise 11) is an orthogonal basis for the subspace $\mathbb{P}_3$ of $\mathbb{P}_4$. Scale the polynomial p_3 so that its vector of values is $(-1, 2, 0, -2, 1)$.
- Let A be any invertible $n \times n$ matrix. Show that for $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$, the formula $\langle \mathbf{u}, \mathbf{v} \rangle = (A\mathbf{u}) \cdot (A\mathbf{v}) = (A\mathbf{u})^T (A\mathbf{v})$ defines an inner product on $\mathbb{R}^n$.
- Let T be a one-to-one linear transformation from a vector space V into $\mathbb{R}^n$. Show that for $\mathbf{u}, \mathbf{v}$ in V , the formula $\langle \mathbf{u}, \mathbf{v} \rangle = T(\mathbf{u}) \cdot T(\mathbf{v})$ defines an inner product on V .

Use the inner product axioms and other results of this section to verify the statements in Exercises 15–18.

- $\langle \mathbf{u}, c\mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$ for all scalars c .

- If $\{\mathbf{u}, \mathbf{v}\}$ is an orthonormal set in V , then $\|\mathbf{u} - \mathbf{v}\| = \sqrt{2}$.
- $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\|\mathbf{u} + \mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{u} - \mathbf{v}\|^2$.
- $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$.

In Exercises 19–24, $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors. Mark each statement True or False (T/F). Justify each answer.

- (T/F) If $\langle \mathbf{u}, \mathbf{u} \rangle = 0$, then $\mathbf{u} = \mathbf{0}$.
- (T/F) If $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, then either $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$.
- (T/F) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle$.
- (T/F) $\langle c\mathbf{u}, c\mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$.
- (T/F) $|\langle \mathbf{u}, \mathbf{u} \rangle| = \langle \mathbf{u}, \mathbf{u} \rangle$.
- (T/F) $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$.
- Given $a \geq 0$ and $b \geq 0$, let $\mathbf{u} = \begin{bmatrix} \sqrt{a} \\ \sqrt{b} \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} \sqrt{b} \\ \sqrt{a} \end{bmatrix}$. Use the Cauchy–Schwarz inequality to compare the geometric mean $\sqrt{ab}$ with the arithmetic mean $(a + b)/2$.
- Let $\mathbf{u} = \begin{bmatrix} a \\ b \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Use the Cauchy–Schwarz inequality to show that

$$\left(\frac{a+b}{2}\right)^2 \leq \frac{a^2+b^2}{2}$$

Exercises 27–30 refer to $V = C[0, 1]$, with the inner product given by an integral, as in Example 7.

- Compute $\langle f, g \rangle$, where $f(t) = 1 - 3t^2$ and $g(t) = t - t^3$.
- Compute $\langle f, g \rangle$, where $f(t) = 5t - 3$ and $g(t) = t^3 - t^2$.
- Compute $\|f\|$ for f in Exercise 27.
- Compute $\|g\|$ for g in Exercise 28.
- Let V be the space $C[-1, 1]$ with the inner product of Example 7. Find an orthogonal basis for the subspace spanned by the polynomials $1, t$, and t^2 . The polynomials in this basis are called *Legendre polynomials*.
- Let V be the space $C[-2, 2]$ with the inner product of Example 7. Find an orthogonal basis for the subspace spanned by the polynomials $1, t$, and t^2 .
- Let $\mathbb{P}_4$ have the inner product as in Example 5, and let p_0, p_1, p_2 be the orthogonal polynomials from that example. Using your matrix program, apply the Gram–Schmidt process to the set $\{p_0, p_1, p_2, t^3, t^4\}$ to create an orthogonal basis for $\mathbb{P}_4$.
- Let V be the space $C[0, 2\pi]$ with the inner product of Example 7. Use the Gram–Schmidt process to create an orthogonal basis for the subspace spanned by $\{1, \cos t, \cos^2 t, \cos^3 t\}$. Use a matrix program or computational program to compute the appropriate definite integrals.